String manipulation

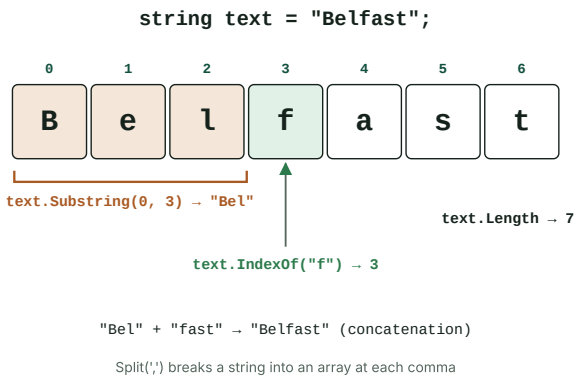
D18 FACT FILE

Specification: students should be able to

- String manipulation functions, including splitting, concatenating, character searching and substring searching

CCEA fact file: Digital Data 2, Programming Constructs 2

Key words: string, concatenation, length, index, substring, Substring, Split, character search, upper, lower



String indexes, Substring, IndexOf and Length for the word Belfast

Key string operations

Concatenation joins strings (with + in C#). The Length property gives the number of characters. Indexing (text[0]) accesses one character and Substring(start, length) extracts part of a string. Searching checks whether a character or substring is present (Contains) or finds its position (IndexOf, -1 if absent). Split(',') breaks a string into an array at a separator such as a comma, which is how CSV records are read.

T Key terms from the mark schemes

Concatenation
Joining strings together.
Credited: U4 2022 Q4

String function
A function that manipulates text, e.g. length, split, find, substring.
Credited: U4 2022 Q4, 2023 Q5

C C# example

String basics

```
string name = "Enniskillen";
Console.WriteLine(name.Length);              // 11
Console.WriteLine(name[0]);                  // E
Console.WriteLine(name[name.Length - 1]);    // n (last character)
```

Exam focus

Name and use string functions: 2022 Q4(c) (join two messages, count characters), 2023 Q5(b) (two functions for password validation). Questions often combine string functions with validation checks.

Unit 4 2023 Q5	[8]	Length, presence and format checks; string functions for a password
Unit 4 2022 Q4	[17]	Tiered pricing algorithm; validation loop 1–50; data types; string functions

? Check yourself

1. What is meant by concatenation? [1]
2. Name a function that returns the number of characters in a string and give an example. [2]
3. word ← 'COMPUTER'. What is the substring from index 3 to index 5 inclusive? [1]

Answers: 1. Joining two or more strings together into one (with + in C#) (1) 2. Length (C#: text.Length) (1) / e.g. "Enniskillen".Length returns 11 (1) 3. "PUT" (C O M P U T E R indexed 0–7: index 3 = P, 4 = U, 5 = T) (1)